

Lab 02

Arithmetic Operations, Discrete-time and Continuous-time Signals and Processing

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task 1

a)

```
x = 15.5 + 3.175i;
```

```
real(x)
```

```
ans =  
15.5000
```

```
imag(x)
```

```
ans =  
3.1750
```

```
conj(x)
```

```
ans =  
15.5000 - 3.1750i
```

```
abs(x)
```

```
ans =  
15.8218
```

```
angle(x)
```

```
ans =  
0.2020
```

b)

```
[a,b] = pol2cart(pi/4,5);  
y = a+(b)*1i;  
real(y)
```

```
ans =  
3.5355
```

```
imag(y)
```

```
ans =  
3.5355
```

```
conj(y)
```

```
ans =  
3.5355 - 3.5355i
```

```
abs(y)
```

```
ans =  
5
```

```
angle(y)
```

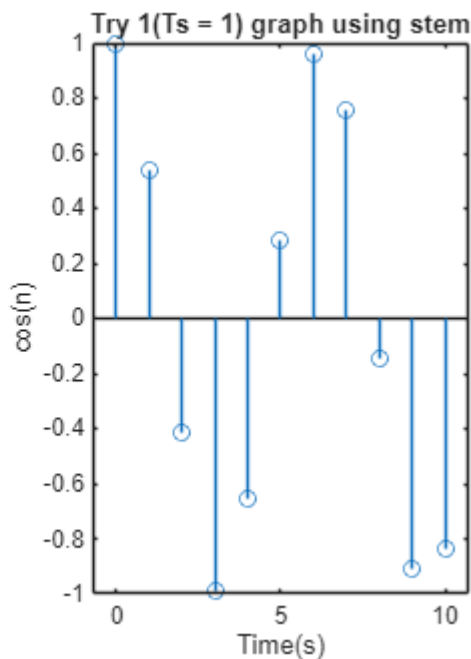
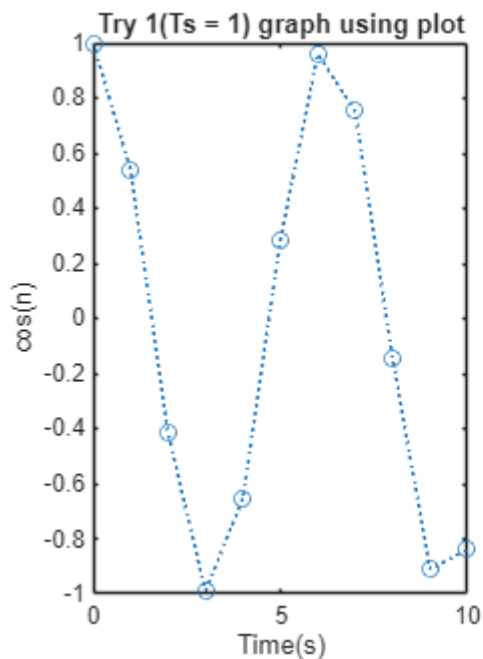
```
ans =  
0.7854
```

task 2

a)

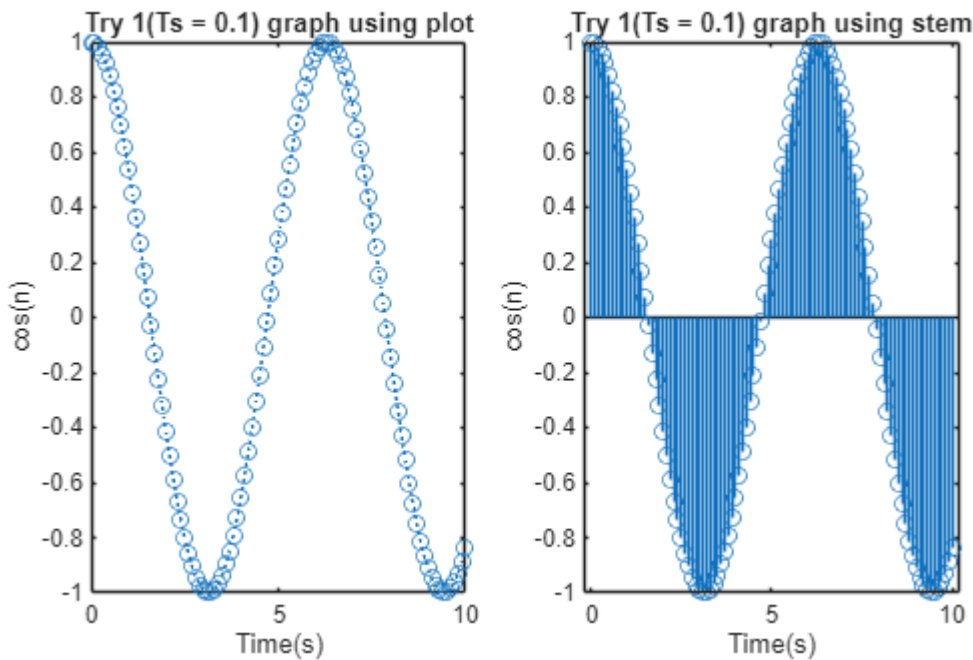
Try 1

```
n = 0:10; % Discrete-time n is defined which goes from 0 to 10 in unitysampling  
interval.  
y = cos(n); % A sinusoidal function y[n] is defined attaining values  $-\pi \leq \angle[\pi] \leq \pi$ .  
subplot 121; plot(n, y, 'o'); % A continuous-time graphical plot of the signal  
y[n] where circular markers are used to represent values connected with a dotted  
line.  
title('Try 1(Ts = 1) graph using plot');  
xlabel('Time(s)'); ylabel('cos(n)');  
subplot 122; stem(n, y); % A more appropriate discrete-time graphical  
representation.  
title('Try 1(Ts = 1) graph using stem');  
xlabel('Time(s)'); ylabel('cos(n)');
```



Try 2

```
Ts = 0.1; % Defining sampling time, the time interval between two consecutive samples
Fs = 1/Ts; % Defining sampling frequency, the number of samples per second
t = 0:Ts:10; % Defining time span
y = cos(t); % Obtaining a discrete-time signal
figure; % Using subplots to depict two plots in one figure
subplot 121; plot(t, y, 'o');
title('Try 1(Ts = 0.1) graph using plot');
xlabel('Time(s)'); ylabel('cos(n)');
subplot 122; stem(t, y);
title('Try 1(Ts = 0.1) graph using stem');
xlabel('Time(s)'); ylabel('cos(n)');
```

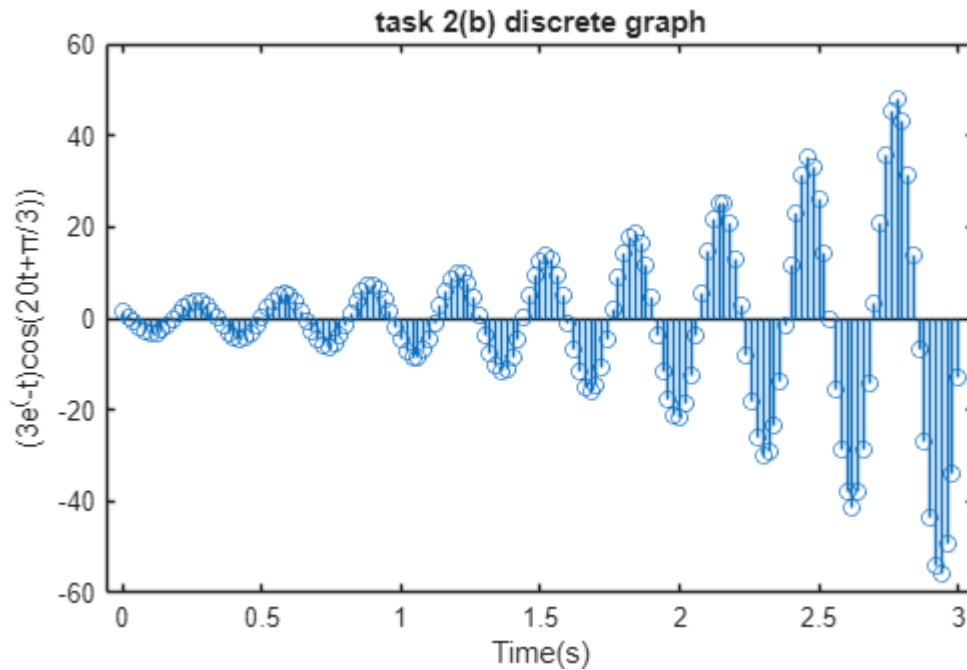


As we make the sampling time smaller **we got better and more clear image** of the sinusoidal graph

the difference between plot and stem was that in **Plot** it makes a continuous graph and then mark a point after each interval of T_s and in **Stem** it only a vertical like of the length of Y at after each interval of T_s

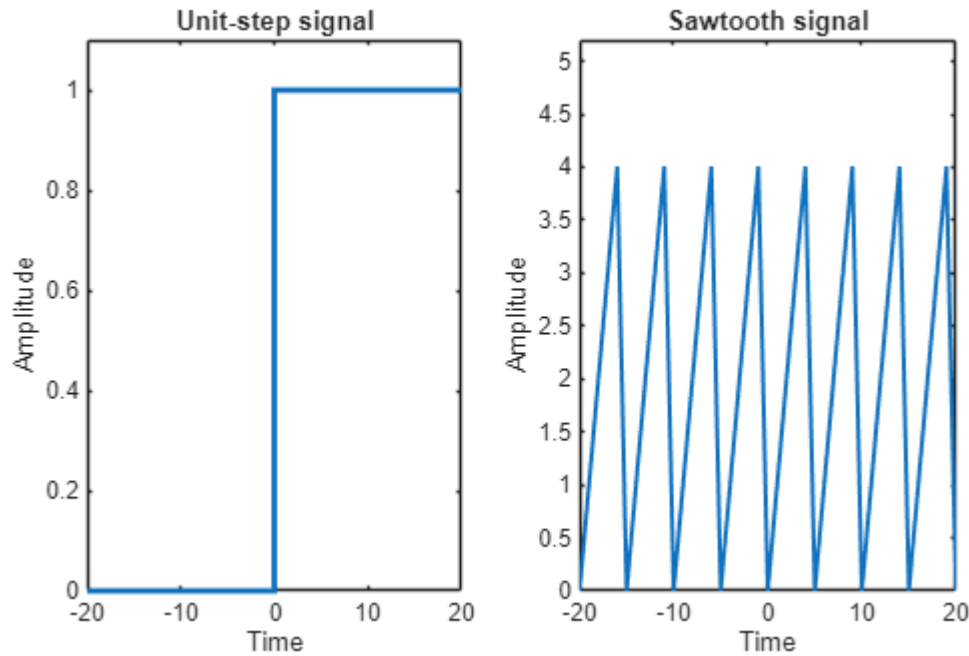
b)

```
Fs = 50;
Ts = 1/Fs;
t = 0:Ts:3;
x = 3.*exp(t).*cos(20.*t+pi/3);
figure;
stem (t, x);
title('task 2(b) discrete graph');
xlabel('Time(s)'); ylabel('(3e^(-t)cos(20t+π/3))');
```



Try 3

```
t = -20:1:20;
y = mod(t,5); %the time vector is divided by 5 and remainder is used to plot the
values resulting in a sawtooth wave when plotted w.r.t time, whereas the second
argument 5 represents the time period/duration of one sawtooth pulse
z = (t>=0); % t>=0 returns logic '1' or '0'
subplot(1,2,1);
stairs(t,z,'linewidth',2)
title('Unit-step signal');
xlabel('Time');
ylabel('Amplitude');
axis([-20 20 0 1.1]);
subplot(1,2,2);
plot(t,y,'linewidth',1.5)
title('Sawtooth signal');
xlabel('Time');
ylabel('Amplitude');
axis([-20 20 0 5.2])
```



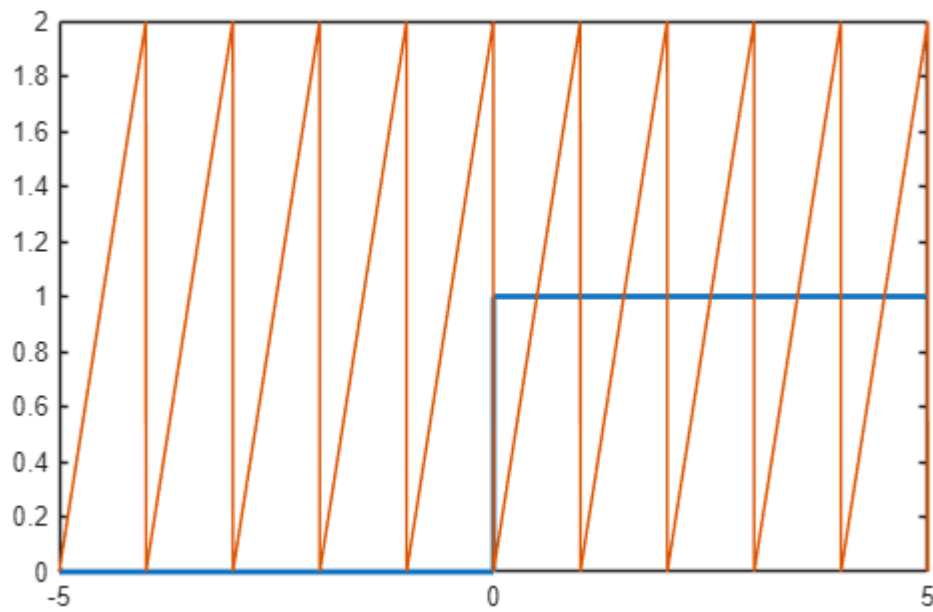
mod(t,x) : it divided the t by x and then return the remainder of that

stairs(t,z,'linewidth',2): it take a time array t and then take a binary array consisting of 1 and 0 then for the graph z(t) whenever value of z be high it will make a vertical step at that point t

axis(a,b,c,d): is used to limit the axis of graph where x axis would be limited from a to b and y axis would be limited from c to d

task 3

```
clc;clear
figure;
t = -20:1/1000:20;
z = (t>=0);
stairs(t,z,'linewidth',2)
hold on
x = sawtooth(2*pi*1*t,1)+1;
plot(t,x)
axis([-5 5 0 2]);
hold off
```



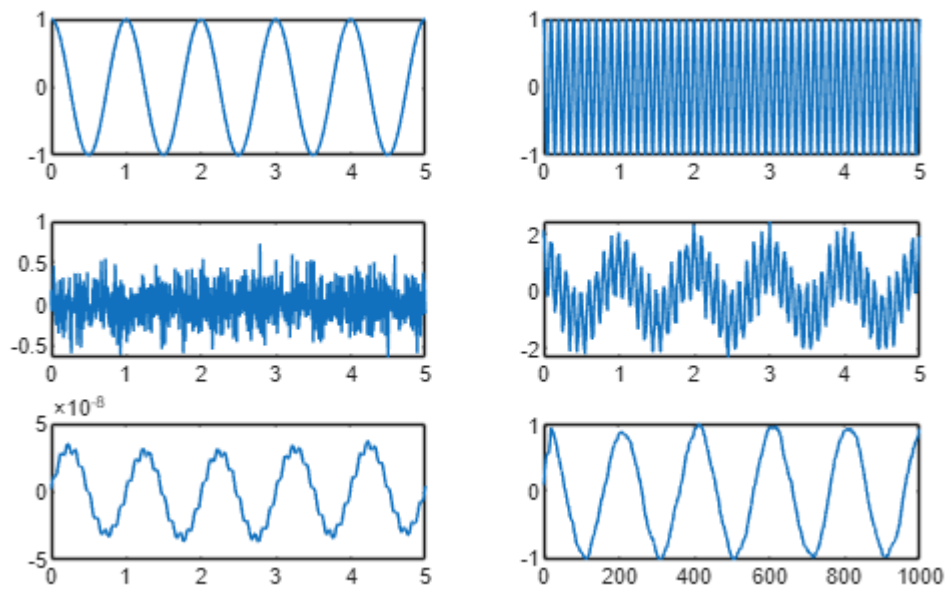
task 4

```

t = 0:0.005:5;
xt1 = cos(2*pi.*t);
xt2 = cos(20*pi.*t);
l = length(xt2);
noise = randn(1,l);
yt = xt1 + xt2 + 0.2.*noise;

subplot (3,2,1);
plot(t,xt1)
subplot (3,2,2);
plot(t,xt2)
subplot(3,2,3);
plot(t,0.2*noise)
subplot(3,2,4);
plot(t,yt)
subplot(3,2,5);
zt = filter(ones(1,5)/5,1,yt);
plot(t,zt)
subplot(3,2, 6);
zt = filter(ones(1,20)/20,1,yt);
plot(zt)

```



so we first added the the xt1 xt2 and noise signals together then we applied the builtin filter funtion inside the matlab and put sample size to be 5 but the sample size was too small so the wave didnt filter correctly thats why we increased the filter size to be 20 at which the wave filtered correctly